

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
 - Determining each option's true quality costs decision-maker time and effort
 - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
 - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
 - Graduate programs invest time and money into students' development
 - Programs may be undervaluing applicants with certain characteristics
 - Future such applicants may receive currently-withheld opportunities

Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
 - Goal is to accept applicants with highest success probabilities who will matriculate
 - Per literature, measure success with program completion and job placement ranking
 - Predictors of acceptance only (success only) may be overweighted (underweighted)
 - ② If they are making mistakes, how can they make better admissions decisions?
 - In turn, by changing decision rules, how much better of a cohort can they admit?
 - Look for applicants who were rejected, would have matriculated, and succeeded
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Methods and Results

- Build structural models of admissions committee's decision problem that support:
 - Statistical tests of whether committee's decision rule aligns with desired objective
 - Counterfactual simulations quantifying gain from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
 - For most applicants, estimated success probabilities align with committee's decisions
 - Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files

Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
 - Krueger & Wu (2000); Athey, Katz, Krueger, Levitt & Poterba (2007); Stock, Finegan & Siegfried (2006–2015); Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
 - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
 - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
 - **Optimality:** Cho & Rust (2010); Hoffman, Kahn & Li (2018); Kleinberg, Lakkaraju, Leskovec, Ludwig & Mullainathan (2018); Anderson, Rosen, Rust & Wong (2025)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which assess optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

① Administrative office receives data from university departments

- Algorithmically extracts variables from textual files, such as recommendation letters

② Synthetic data (Patki et al. 2016) used for model development

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with < 5 comparable observations binned to prevent attribute disclosure

③ Administrative office executes final estimation code on actual data

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality and Gains

Data Description

- **Sample:** Applicants to research university's economics PhD program from 2015–2019
- **Outcomes:** Measures of admission and success
 - **Admission:** Read File $\Rightarrow$ Accept $\Rightarrow$ Matriculate $\Rightarrow$ Success
 - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Covariates:** Objective vs. subjective variables
 - **Objective:** Demographics and performance measures, such as GRE scores
 - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
 - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
 - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
 - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
 - **Success:** *PhD Program Rank* is best predictor. Rank held equal, other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

» Insufficiency of Comparing AMEs

» Diversity Objective

» Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to assess optimality
- Need model of admissions to relate acceptance effects to success effects

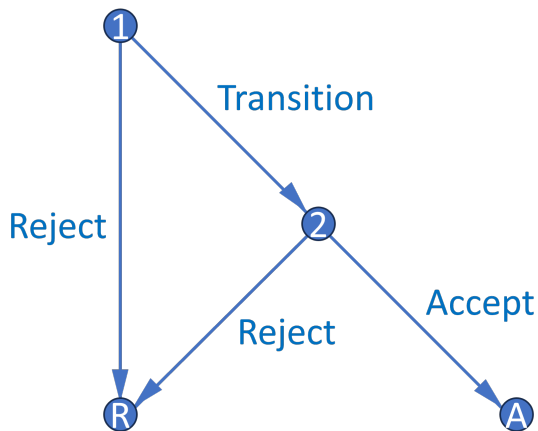
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- 2 Structural Models**
- 3 Structural Estimation
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Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
 - Maximize admitted cohort's quality, considering matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
 - Quantify gains from deviating from actual to counterfactual decision rule

Dynamics



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

Portfolio Choice Problem

For applicant n , choose $d_{1,n} \in \{\text{Transition, Reject}\}$ and $d_{2,n} \in \{\text{Accept, Reject}\}$:

Round 2:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

Round 1:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{\frac{EV_{2,n}(x_n)}{\mathbb{E}[\text{Round 2 Value}]}}_{\text{Transition}} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success, Failure}\}$,
 N = number of applicants, (N_T, N_A) = max number of files read/acceptances

Portfolio Choice to Cutoff Rule

- Assuming no complementarities, portfolio choice problem has cutoff rule solution
 - Round 2:** Accept transitioned applicant if success probability exceeds Round 2 cutoff
 - Round 1:** Transition applicant if Round 2 expected value exceeds Round 1 cutoff
 - Decisions across applicants independent, so no need to consider all subsets

Round 2:

$$d_{2,n}^* = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \text{Cutoff}_2 \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \text{Cutoff}_2 \\ R & \text{if } P(y_n = S | x_n, z_n) < \text{Cutoff}_2 \end{cases}$$

Round 1:

$$d_{1,n}^* = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \text{Cutoff}_1 \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \text{Cutoff}_1 \\ R & \text{if } EV_{2,n}(x_n) < \text{Cutoff}_1 \end{cases}$$

Cutoff Rule Model

- ★ **Goal:** Use choices and outcomes data to estimate success probabilities, while respecting constraint on number of files committee can read
- ★ **Legend:** Estimated model objects vs. Choices and outcomes data

Cutoff Rule Model, Round 2

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Cutoff Rule Model, Round 1

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, (y_1^*, y_2^*) = marginal outcomes

Conditional Choice Probabilities

Round 2:

$$P(d_2 = A | x, z, t, \bar{k}) = \frac{\exp(u_2(\cdot, A)/\sigma)}{\exp(u_2(\cdot, A)/\sigma) + \exp(u_2(\cdot, R)/\sigma)}$$

Round 1:

$$P(d_1 = T | x, t, \bar{k}) = \frac{\exp(u_1(\cdot, T)/\sigma)}{\exp(u_1(\cdot, T)/\sigma) + \exp(u_1(\cdot, R)/\sigma)}$$

» Dynamic Parameterization

» Static Model

Portfolio Choice with Matriculation

Round 2:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^{N_T} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$

Round 1:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where (x, z) = variables, $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$,
 N = number of applicants, (N_T, N_M) = max number of files read/matriculants

Cutoff Rule with Matriculation

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}},$$

such that: $\sum_{n=1}^{N_T} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_M,$

Theorem 1

*For large N_T , Round 2 problem has a cutoff rule solution on **success** probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- Matriculation probabilities matter in Round 1, especially for low-ranked programs

Cutoff Rule with Matriculation, Round 2

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get Round 2 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Cutoff Rule with Matriculation, Round 1

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get Round 1 Cutoff}}$$

- **States:** (x, z) = objective and subjective variables, t = year, $\bar{k}$ = fixed program rank, (ϵ_1, ϵ_2) = Type 1 extreme value preference shocks with scale parameter σ
- **Choices:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $m \in \{\text{Matriculate, Decline}\}$, (y_1^*, y_2^*, m_1^*) = marginal outcomes

Importance of Matriculation

$$\lim_{\sigma \rightarrow 0} EV_2 = \mathbb{E}_{\tilde{z}}[\max\{P(y = S|x, \tilde{z}, t, \bar{k}) - P(y_2^* = S|t), 0\}P(m = M|x, \tilde{z}, t)|x, t, \bar{k}]$$

- Matriculation matters when expected success probability $\gg$ Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

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Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$
$$\prod_{n=1}^N \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)} (1 - P_{d_{1,n}}(\theta_T))^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$ aligns success probabilities implied by outcomes with those implied by CCPs

Cutoff Probability Identification

- Identify each year's cutoffs by how selective committee is that year
 - Fewer acceptances and stronger applicants increase cutoff
 - More acceptances and weaker applicants decrease cutoffs
- While selectivity identifies cutoffs, sorting improves restricted likelihood

Theorem 2

Cutoff probability estimate is such that number of acceptances N_A equals expected N_A . For any cutoff and N_A , likelihood is highest when committee accepts best N_A applicants.

Restricted Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.3731 (0.3067, 0.4539)	36.58% (29.81%, 43.92%)	
⋮	⋮	⋮	
2019	0.3887 (0.3268, 0.4624)	38.14% (31.89%, 44.81%)	
Sigma			0.0020 (0.0013)

- Round 1 cutoffs relative to EV_2 . Round 2 cutoffs relative to success probabilities
- Cutoffs stable across years. $\hat{\sigma}$ small $\Leftrightarrow$ distorted restricted success probabilities

Restricted Estimates with Matriculation

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
⋮	⋮	⋮	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)

- With matriculation, EV_2 's functional form yields lower Round 1 cutoffs

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Deviation Gains

- Suppose committee adopted model's decision rule with N_T unchanged for now
- Would average success probability of acceptance/matriculation set have increased?
 - 1 Select applicants with model, and determine its acceptance/matriculation set
 - 2 Compute average success probability of acceptance/matriculation set
 - 3 Compare average success probability to that of committee's decision rule's set
- Plug $\hat{\theta}_S^U$ into model because $\hat{\theta}_S^R$ may be distorted to rationalize suboptimal behavior

Deviation Gains Test

- Is deviation gain significant? Do z test of $H_0: g(\hat{\theta}_S^U) = 0$ vs. $H_1: g(\hat{\theta}_S^U) > 0$
 - ① Gain is approximately normal, so test statistic is standardized gain. **Or:**
 - ② Compute average success probabilities across B perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$

Theorem 3

P-value is fraction of perturbations with model's $\overline{P(\text{Success})}$ less than committee's.

- Model's set must be exogenously chosen, e.g. train and test on different data
 - If not, gain will be biased upward, and test will be biased toward rejection
 - In bigger dataset, such as hiring at large company, holdout set is feasible

Deviation Gains Correction

- Observed deviation gain is combination of true gain and “winner’s curse”

$$\begin{aligned} g(\hat{\theta}_S^U) &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U)}_{\text{Model}} - \underbrace{\bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{Committee}} \\ &= \underbrace{\bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U)}_{\text{Winner's curse}} + \underbrace{\bar{P}_{\mathcal{A}^R(\theta_S^0)}(\hat{\theta}_S^U) - \bar{P}_{\mathcal{A}^U}(\hat{\theta}_S^U)}_{\text{True gain at } \hat{\theta}_S^U} \end{aligned}$$

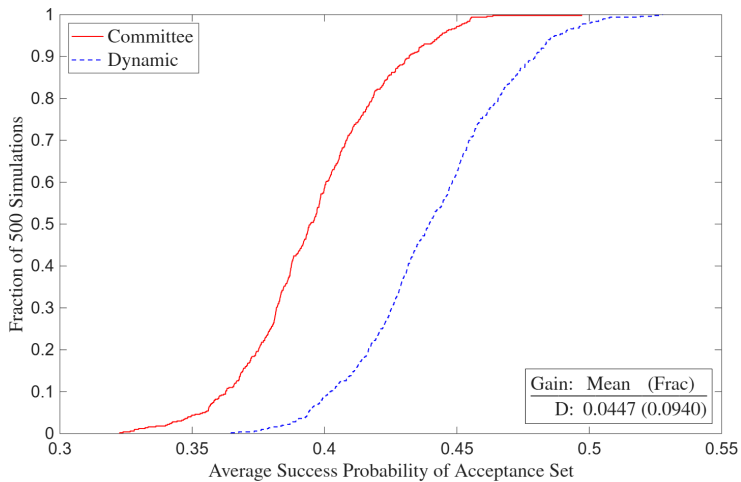
- Subtract estimate of winner’s curse from observed deviation gain

$$\widehat{\text{Winner's curse}} = \frac{1}{B} \sum_{b=1}^B [\bar{P}_{\mathcal{A}^R(\tilde{\theta}_{S,b}^U)}(\tilde{\theta}_{S,b}^U) - \bar{P}_{\mathcal{A}^R(\hat{\theta}_S^U)}(\tilde{\theta}_{S,b}^U)]$$

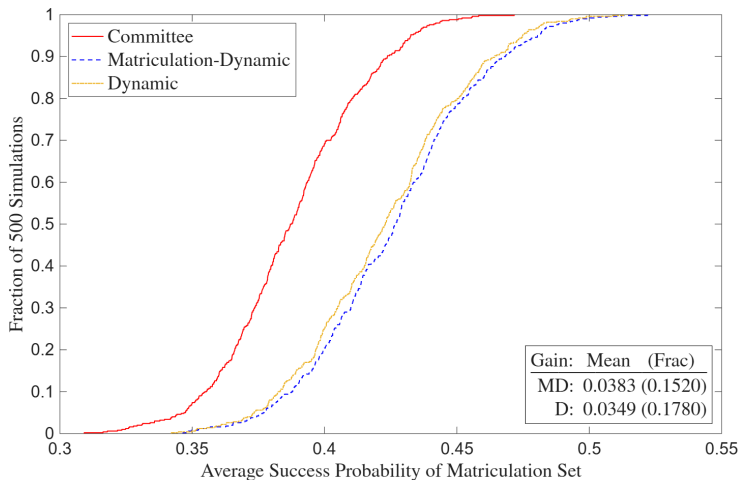
Theorem 4

Correction is exact under certain conditions and removes at least $\sqrt{2} - 1$ of the curse.

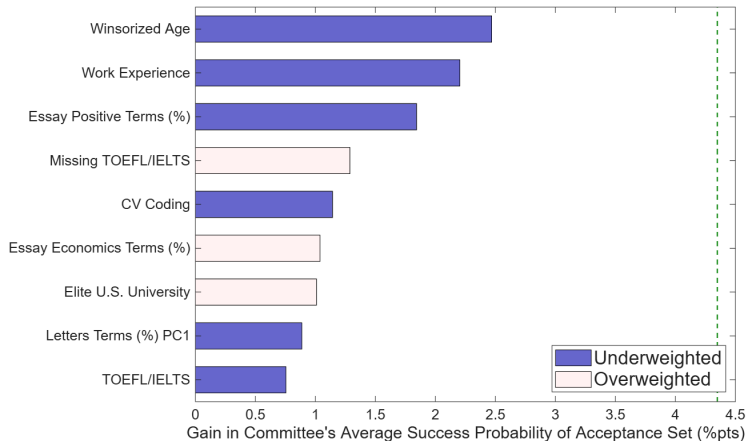
Deviation Gains (Acceptance)



Deviation Gains (Matriculation)



Deviation Gains Attribution (Acceptance)

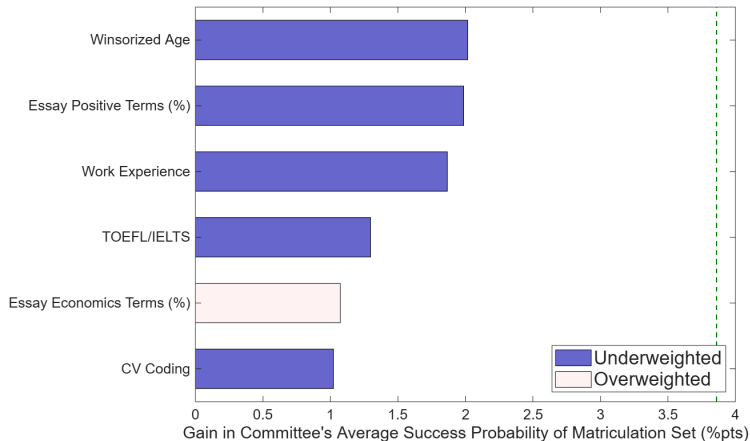


- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Acceptance)

» Deviation Gains Attribution Table (Acceptance)

Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

» Full Deviation Gains Attribution (Matriculation)

» Deviation Gains Attribution Table (Matriculation)

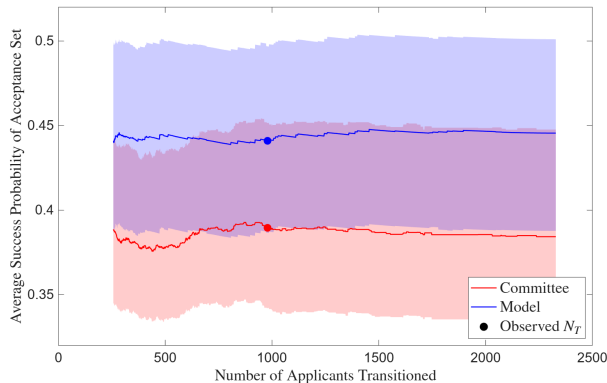
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Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

- Blue = underweighted, red = overweighted, white = neither
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

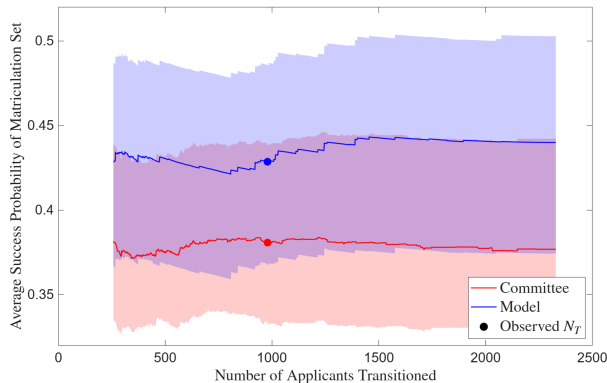
» Acceptance vs. Success AMEs

Benefit of Reading More Files (Acceptance)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - cn$
- **FOC:** $V = c / (N_A \bar{P}'(N_T))$
- V is dollar value of an additional expected success

Benefit of Reading More Files (Matriculation)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_M \bar{P}(n) - cn$
- **FOC:** $V = c / (N_M \bar{P}'(N_T))$
- V is dollar value of an additional expected success

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to assess optimality and quantify deviation gains
 - ① For most applicants, estimated success probabilities align with committee's decisions
 - ② Reweighting some applicant characteristics could have increased admitted cohort's average success probability by 4 percentage points without reading more files
 - ③ Success is hard to predict, but even modest gains in selection are worthwhile
- ★ Methods apply to undergraduate/professional admissions, hiring, and other settings

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	70	1	0	1	0	.	27
0	0	0	0	0	92	1	1	0	1	.	150
0	0	0	0	0	85	1	1	0	0	.	80
0	0	0	0	0	72	1	0	0	1	.	51
0	0	0	0	0	74	0	1	0	0	.	150
0	0	0	0	0	79	0	0	0	1	.	26
1	1	0	0	1	97	0	0	1	1	8	43
0	0	0	0	0	76	1	0	0	0	.	17
1	1	0	0	0	75	0	0	0	0	3	70
0	0	0	0	0	75	0	0	1	0	.	150



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	92	0	0	1	1	8	31
0	0	0	0	0	83	0	0	0	1	.	60
0	0	0	0	1	94	0	1	0	1	.	19
1	1	1	1	0	90	0	0	1	1	5	37
0	0	0	0	1	86	0	0	1	0	.	52
1	0	1	0	1	91	1	1	0	0	2	29
1	1	0	0	0	86	0	0	0	1	5	71
1	1	0	0	0	98	0	1	0	0	5	21
1	1	0	0	1	88	1	1	0	1	4	37
0	0	0	0	0	92	1	1	0	0	.	150

A1: Dependent Variables

Admission	Success
1 (Transition)	1 (Complete Program)
<i>1 (Accept)</i>	<i>1 (Academic Placement)</i>
1 (Waitlist)	1 (Government Placement)
1 (Open House)	1 (Consulting Placement)
<i>1 (Matriculate)</i>	<i>1 (Tech Placement)</i>
1 (Attend Program)	1 (Other Placement)
Program Rank	1 (No Placement)
	Placement Rank
	1 (Top 100 Placement)

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
 - For non-tenure track, use weighted average of university's and maximum rankings
 - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

A1: Objective Variables

Demographic	Performance
Year	GRE Verbal Percentile
1 (<i>Covid Year</i>)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1 (Female)	Undergraduate GPA
1 (U.S.)	TOEFL/IELTS
1 (<i>U.S. African/Hispanic/Native</i>)	1 (Elite U.S. University)
1 (<i>U.S. Asian</i>)	1 (Elite U.S. Liberal Arts College)
1 (<i>U.S. Other</i>)	1 (Other U.S. University/College)
1 (International Asian)	1 (Elite Foreign University)
1 (International Other)	1 (Other Foreign University)
	1 (Graduate Degree)
	1 (Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

A1: Subjective Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1 (Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1 (Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1 (Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Eonic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

A1: Dictionaries

● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
 - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
 - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
 - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
 - If GRE predicts success, committee should use it regardless of why

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
 - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
 - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
 - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
 - **Instruments:** Variables that predict matriculation but not the outcome

A1: Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0037	0.0053*
Female	0.0163**	-0.0490	-0.0157	-0.0032
U.S.	-0.0013	0.1472*	0.0213	0.0192
International Asian	-0.0214***	0.1644**	0.0298	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0028**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.0721	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0219	0.0409*
Work Experience	0.0140**	0.1591**	0.0271	0.0358**
#Prolific Recommenders	0.0039	-0.0426	0.0137*	0.0076
Committee Score	0.0788***	-0.0371	0.0144**	0.0007
Advanced Math	0.0246	-0.1950**	-0.0317	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0038	0.0159
CV Coding	-0.0044*	-0.0414**	0.0108	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0175**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0118*	0.0123*
Missing Committee Score	-0.0096		-0.0528**	-0.0277
Open House		0.2615***		
Program Rank				-0.0011***
Missing Program Rank				-0.3965***

Year Dummies

GRE Verbal Percentile

GRE Analytic Percentile

Elite U.S. University

Elite U.S. Liberal Arts College

Elite Foreign University

Other Foreign University

Graduate Degree

#Professor Recommenders

#Math Courses

Letters Length/100

Letters Terms (%) PC2

CV Length/100

CV Publication/Working Paper

Essay Length/100

Essay Specific Professor

Essay Research Terms (%)

Essay Negative Terms (%)

Missing Dummies

A1: Insufficiency of Comparing AMEs

- Cannot assess optimality by comparing AMEs across regressions. Suppose:
 - Work experience has large effect on success, graduate degree smaller effect
 - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
 - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
 - **Acceptance Regression:** $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
 - **Success Regression:** $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
 - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

A1: Diversity Objective

- **Demographics:** Squares vs. circles

- **Quality:** High vs. low

High Quality:



Low Quality:



A1: Diversity Objective

- No preference for squares or circles

High Quality:



Low Quality:



» Acceptance vs. Success AMEs

A1: Diversity Objective

- Small preference for squares $\Rightarrow$ consistent with maximizing quality

High Quality:



Low Quality:



» Acceptance vs. Success AMEs

A1: Diversity Objective

- Large preference for squares $\Rightarrow$ inconsistent with maximizing quality

High Quality:



Low Quality:



►► Acceptance vs. Success AMEs

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
 - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
 - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
 - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
 - **Problem:** Not a test if admissions decisions are optimal
 - Need structural model of admissions to conclude about optimality

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$,

and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. Then:
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

A1: Linear Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics) $|$ Predictors.

Let $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$, and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\text{Standardized Committee Score, Standardized Placement Rank}\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
Then $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$)

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
 - Need different skills at different stages in the program, such as grades vs. placement
 - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
 - **Admission:** May have different preparation and/or interests
 - **Completion:** Depending on country, may have fewer outside options
 - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
 - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
 - For economics, increased importance of elite predocs and Fed RAs

A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
 - ① **Static Model:** In Round 1, rank applicants based only on objective variables
 - ② **Myopic Model:** Like dynamic, but distribution of z not conditional on x
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \bar{k})$, where Success = $\mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t 's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year t 's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
 - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ = expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
 - **Static:** If transition applicant, get $P(y = S|x, t, \bar{k})$, or $P(\text{Success})$ unconditional on z . It may differ from $P(y = S|x, z, t, \bar{k})$, though it still accounts for z 's effect on y
 - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, replace $F(z|x, t)$ with $F(z|t)$

A2: Lagrangian (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Lagrangian with Shocks

Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 5

For large N , there is a cutoff rule solution with logistic CCPs. ► Portfolio Choice to Cutoff Rule

A2: Lagrangian, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Lagrangian, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

A2: Lagrangian, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

A2: Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S | x, t, \bar{k})$ estimates inconsistent, but fitted values account for z 's effect on y

A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

A2: Lagrangian with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

A2: Lagrangian with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))] P(m = M|w)f(w)dw + \lambda(N_M - \overline{N}_M)$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

A2: Lagrangian with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

A2: Lagrangian with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

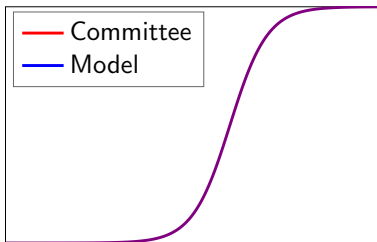
$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}} \\ & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\ & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \end{aligned}$$

A2: Empirical Considerations

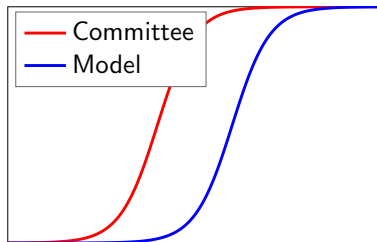
- **Selection Effects:** Track success for all applicants, including non-matriculants
 - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
 - In decision blocks containing success, fix *Program Rank* at this university's rank
 - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
 - **Example:** Recommendation letter has nuances that algorithms can't detect
 - Include subjective committee score in z as a proxy for this information

A2: Deviation Gains Procedure

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 For this comparison, evaluate selections at perturbations of estimates



Optimal Committee



Suboptimal Committee

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \bar{k})$ for static. Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept top N_A survivors.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for static (best analogue). Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of matriculation set, and compare to committee's set

A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
 - Too few observations for a true out-of-sample test, and cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
 - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
 - Comprehensive data, especially committee score in z , protects against this limitation

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
 - **Example:** Applicants with work experience more likely to succeed and matriculate
 - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
 - Instead maximize combination of $P(\text{Success})$ and other objectives
 - **Example:** Committee may want diverse cohort (e.g. across fields):
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see [online appendix](#))
 - Three-round models, and two-round models that repeat over infinite horizon
 - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

A3: Summary Statistics

Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Profic Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

A3: Recommendation Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

A3: Logistic Regression Results

2019	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000	1999	1998	1997	1996	1995	1994	1993	1992	1991	1990	1989	1988	1987	1986	1985	1984	1983	1982	1981	1980	1979	1978	1977	1976	1975	1974	1973	1972	1971	1970	1969	1968	1967	1966	1965	1964	1963	1962	1961	1960	1959	1958	1957	1956	1955	1954	1953	1952	1951	1950	1949	1948	1947	1946	1945	1944	1943	1942	1941	1940	1939	1938	1937	1936	1935	1934	1933	1932	1931	1930	1929	1928	1927	1926	1925	1924	1923	1922	1921	1920	1919	1918	1917	1916	1915	1914	1913	1912	1911	1910	1909	1908	1907	1906	1905	1904	1903	1902	1901	1900	1899	1898	1897	1896	1895	1894	1893	1892	1891	1890	1889	1888	1887	1886	1885	1884	1883	1882	1881	1880	1879	1878	1877	1876	1875	1874	1873	1872	1871	1870	1869	1868	1867	1866	1865	1864	1863	1862	1861	1860	1859	1858	1857	1856	1855	1854	1853	1852	1851	1850	1849	1848	1847	1846	1845	1844	1843	1842	1841	1840	1839	1838	1837	1836	1835	1834	1833	1832	1831	1830	1829	1828	1827	1826	1825	1824	1823	1822	1821	1820	1819	1818	1817	1816	1815	1814	1813	1812	1811	1810	1809	1808	1807	1806	1805	1804	1803	1802	1801	1800	1799	1798	1797	1796	1795	1794	1793	1792	1791	1790	1789	1788	1787	1786	1785	1784	1783	1782	1781	1780	1779	1778	1777	1776	1775	1774	1773	1772	1771	1770	1769	1768	1767	1766	1765	1764	1763	1762	1761	1760	1759	1758	1757	1756	1755	1754	1753	1752	1751	1750	1749	1748	1747	1746	1745	1744	1743	1742	1741	1740	1739	1738	1737	1736	1735	1734	1733	1732	1731	1730	1729	1728	1727	1726	1725	1724	1723	1722	1721	1720	1719	1718	1717	1716	1715	1714	1713	1712	1711	1710	1709	1708	1707	1706	1705	1704	1703	1702	1701	1700	1699	1698	1697	1696	1695	1694	1693	1692	1691	1690	1689	1688	1687	1686	1685	1684	1683	1682	1681	1680	1679	1678	1677	1676	1675	1674	1673	1672	1671	1670	1669	1668	1667	1666	1665	1664	1663	1662	1661	1660	1659	1658	1657	1656	1655	1654	1653	1652	1651	1650	1649	1648	1647	1646	1645	1644	1643	1642	1641	1640	1639	1638	1637	1636	1635	1634	1633	1632	1631	1630	1629	1628	1627	1626	1625	1624	1623	1622	1621	1620	1619	1618	1617	1616	1615	1614	1613	1612	1611	1610	1609	1608	1607	1606	1605	1604	1603	1602	1601	1600	1599	1598	1597	1596	1595	1594	1593	1592	1591	1590	1589	1588	1587	1586	1585	1584	1583	1582	1581	1580	1579	1578	1577	1576	1575	1574	1573	1572	1571	1570	1569	1568	1567	1566	1565	1564	1563	1562	1561	1560	1559	1558	1557	1556	1555	1554	1553	1552	1551	1550	1549	1548	1547	1546	1545	1544	1543	1542	1541	1540	1539	1538	1537	1536	1535	1534	1533	1532	1531	1530	1529	1528	1527	1526	1525	1524	1523	1522	1521	1520	1519	1518	1517	1516	1515	1514	1513	1512	1511	1510	1509	1508	1507	1506	1505	1504	1503	1502	1501	1500	1499	1498	1497	1496	1495	1494	1493	1492	1491	1490	1489	1488	1487	1486	1485	1484	1483	1482	1481	1480	1479	1478	1477	1476	1475	1474	1473	1472	1471	1470	1469	1468	1467	1466	1465	1464	1463	1462	1461	1460	1459	1458	1457	1456	1455	1454	1453	1452	1451	1450	1449	1448	1447	1446	1445	1444	1443	1442	1441	1440	1439	1438	1437	1436	1435	1434	1433	1432	1431	1430	1429	1428	1427	1426	1425	1424	1423	1422	1421	1420	1419	1418	1417	1416	1415	1414	1413	1412	1411	1410	1409	1408	1407	1406	1405	1404	1403	1402	1401	1400	1399	1398	1397	1396	1395	1394	1393	1392	1391	1390	1389	1388	1387	1386	1385	1384	1383	1382	1381	1380	1379	1378	1377	1376	1375	1374	1373	1372	1371	1370	1369	1368	1367	1366	1365	1364	1363	1362	1361	1360	1359	1358	1357	1356	1355	1354	1353	1352	1351	1350	1349	1348	1347	1346	1345	1344	1343	1342	1341	1340	1339	1338	1337	1336	1335	1334	1333	1332	1331	1330	1329	1328	1327	1326	1325	1324	1323	1322	1321	1320	1319	1318	1317	1316	1315	1314	1313	1312	1311	1310	1309	1308	1307	1306	1305	1304	1303	1302	1301	1300	1299	1298	1297	1296	1295	1294	1293	1292	1291	1290	1289	1288	1287	1286	1285	1284	1283	1282	1281	1280	1279	1278	1277	1276	1275	1274	1273	1272	1271	1270	1269	1268	1267	1266	1265	1264	1263	1262	1261	1260	1259	1258	1257	1256	1255	1254	1253	1252	1251	1250	1249	1248	1247	1246	1245	1244	1243	1242	1241	1240	1239	1238	1237	1236	1235	1234	1233	1232	1231	1230	1229	1228	1227	1226	1225	1224	1223	1222	1221	1220	1219	1218	1217	1216	1215	1214	1213	1212	1211	1210	1209	1208	1207	1206	1205	1204	1203	1202	1201	1200	1199	1198	1197	1196	1195	1194	1193	1192	1191	1190	1189	1188	1187	1186	1185	1184	1183	1182	1181	1180	1179	1178	1177	1176	1175	1174	1173	1172	1171	1170	1169	1168	1167	1166	1165	1164	1163	1162	1161	1160	1159	1158	1157	1156	1155	1154	1153	1152	1151	1150	1149	1148	1147	1146	1145	1144	1143	1142	1141	1140	1139	1138	1137	1136	1135	1134	1133	1132	1131	1130	1129	1128	1127	1126	1125	1124	1123	1122	1121	1120	1119	1118	1117	1116	1115	1114	1113	1112	1111	1110	1109	1108	1107	1106	1105	1104	1103	1102	1101	1100	1099	1098	1097	1096	1095	1094	1093	1092	1091	1090	1089	1088	1087	1086	1085	1084	1083	1082	1081	1080	1079	1078	1077	1076	1075	1074	1073	1072	1071	1070	1069	1068	1067	1066	1065	1064	1063	1062	1061	1060	1059	1058	1057	1056	1055	1054	1053	1052	1051	1050	1049	1048	1047	1046	1045	1044	1043	1042	1041	1040	1039	1038	1037	1036	1035	1034	1033	1032	1031	1030	1029	1028	1027	1026	1025	1024	1023	1022	1021	1020	1019	1018	1017	1016	1015	1014	1013	1012	1011	1010	1009	1008	1007	1006	1005	1004	1003	1002	1001	1000	999	998	997	996	995	994	993	992	991	990	989	988	987	986	985	984	983	982	981	980	979	978	977	976	975	974	973	972	971	970	969	968	967	966	965	964	963	962	961	960	959	958	957	956	955	954	953	952	951	950	949	948	947	946	945	944	943	942	941	940	939	938	937	936	935	934	933	932	931	930	929	928	927	926	925	924	923	922	921	920	919	918	917	916	915	914	913	912	911	910	909	908	907	906	905	904	903	902	901	900	899	898	897	896	895	894	893	892	891	890	889	888	887	886	885	884	883	882	881	880	879	878	877	876	875	874	873	872	871	870	869	868	867	866	865	864	863	862	861	860	859	858	857	856	855	854	853	852	851	850	849	848	847	846	845	844	843	842	841	840	839	838	837	836	835	834	833	832	831	830	829	828	827	826	825	824	823	822	821	820	819	818	817	816	815	814	813	812	811	810	809	808	807	806	805	804	803	802	801	800	799	798	797	796	795	794	793	792	791	790	789	788	787	786	785	784	783	782	781	780	779	778	777	776	775	774	773	772	771	770	769	768	767	766	765	764	763	762	761	760	759	758	757	756	755	754	753	752	751	750	749	748	747	746	745	744	743	742	741	740	739	738	737	736	735	734	733	732	731	730	729	728	727	726	725	724	723	722	721	720	719	718	717	716	715	714	713	712	711	710	709	708	707	706	705	704	703	702	701	700	699	698	697	696	695	694	693	692	691	690	689	688	687	686	685	684	683	682	681	680	679	678	677	676	675	674	673	672	671	670	669	668	667	666	665	664	663	662	661	660	659	658	657	656	655	654	653	652	651	650	649	648	647	646	645	644	643	642	641	640	639	638	637	636	635	634	633	632	631	630	629	628	627	626	625	624	623	622	621	620	619	618	617	616	615	614	613	612	611	610	609	608	607	606	605	604	603	602	601	600
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- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs

[illegible]

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression Results (Synthetic)

[illegible]

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Logistic Regression AMEs (Synthetic)

[illegible]

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2 (-), Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A3: Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017	-0.4928								
2018	-2.4456	0.0186							
2019	-1.7752								
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959						0.0477		
International Asian	-0.8686	-0.8651	0.1271			-0.5295	-0.7927		
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015		-0.0085			
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125	-0.2706	-0.2243	-0.0070	
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316		0.2561	
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. Liberal Arts College	0.4024	0.1112							
Elite Foreign University	-0.4025								
Other Foreign University	0.0790	0.0940	0.1998					-0.2265	
Missing Undergraduate Institution		-0.0036							
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976		-2.0809	0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295	1.8402	2.2548	-0.4899	
Committee Score		3.4243			0.0353				
#Math Courses	-0.0553	0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014		-0.0286	-0.1179		
CV Coding	-0.0038	-0.1754		0.0165	0.0076	-0.2540	-0.4154	-0.0998	
CV Publication/Working Paper	0.5563	0.6436		0.0892		-0.0687			
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	0.4473	0.6570	0.2117	
Essay Research Terms (%)		-0.0278					-0.1506		
Essay Positive Terms (%)	-0.0049	0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714	4.0146	0.7071	
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082		0.1385	
Missing Program Rank						-2.2557		19.4273	
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R ²	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R ²	0.2441	0.7605	0.0596	0.0062	0.0129	0.2190	0.0121	0.0183	0.2638

- Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

Restricted

[illegible]

A3: Restricted Static Estimates

	Round 1 Cutoff	Round 2 Cutoff	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
⋮	⋮	⋮	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064 (0.0035)

- Static model has interpretable Round 1 cutoff probabilities
- Round 2 cutoff probabilities always greater than those in Round 1

A3: Myopic Estimates

Unrestricted

[illegible]

	(0.001)	(0.001)	(0.001)	(0.001)
Obs./Param. LL AIC	2120	207	10470.7	20335.6

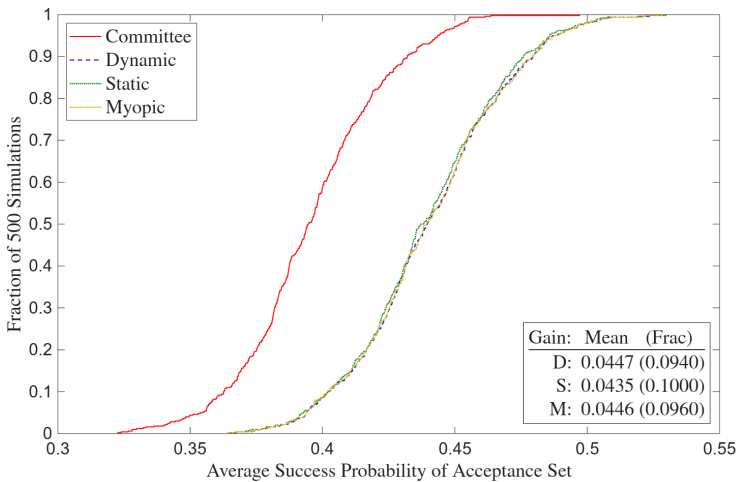
Robust standard errors in parentheses. $\chi^2_{0.10}$ 19.33, $\chi^2_{0.05}$ 22.36, $\chi^2_{0.01}$ 38.58

Restricted

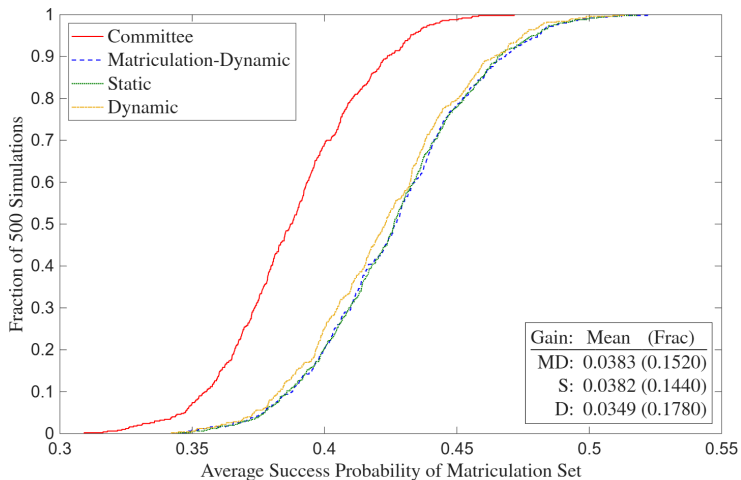
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Robust standard errors in parentheses. * $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$. Cellphone controls are 1970.

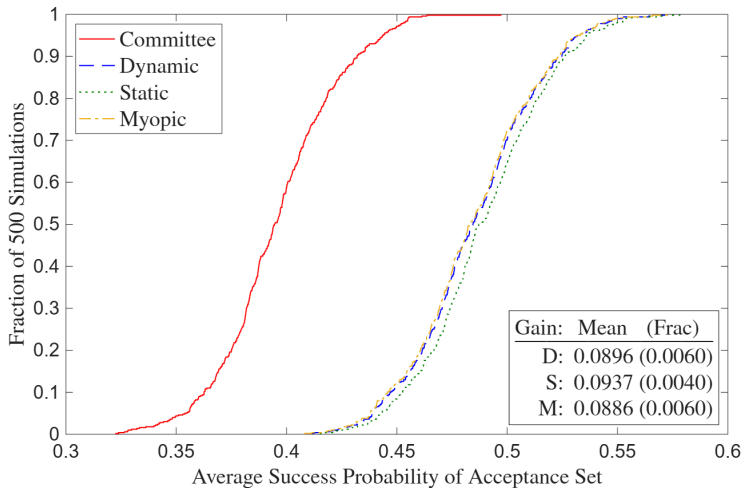
A3: Full Deviation Gains (Acceptance)



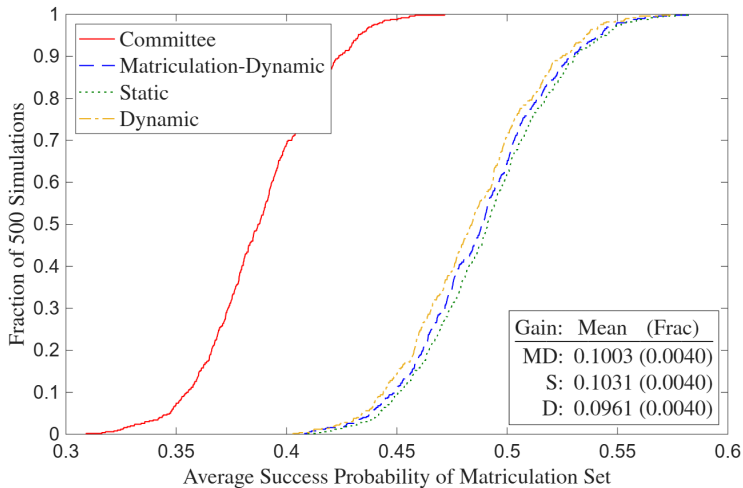
A3: Full Deviation Gains (Matriculation)



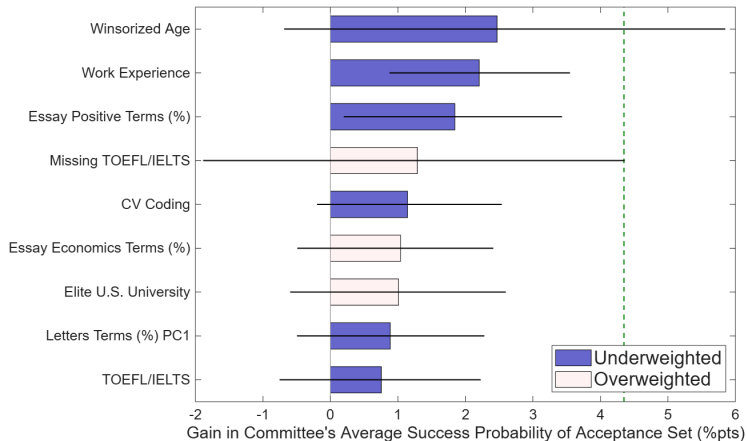
A3: Uncorrected Deviation Gains (Acceptance)



A3: Uncorrected Deviation Gains (Matriculation)

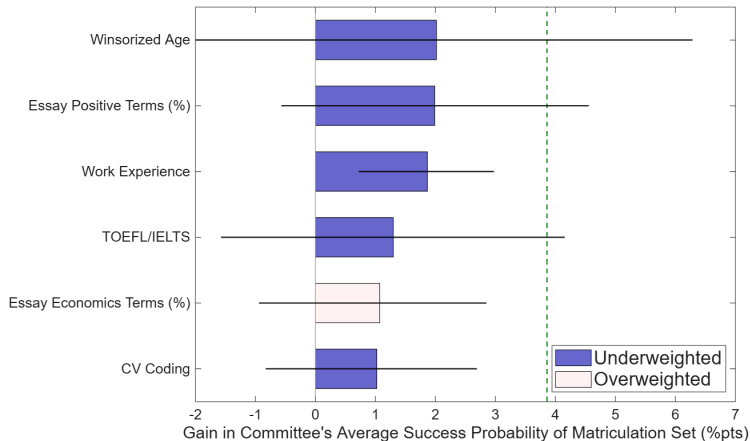


A3: Full Deviation Gains Attribution (Acceptance)



- ① Experience
- ② English Proficiency
- ③ Essay/Letters Subtext
- ④ Industry Background
- ⑤ Undergraduate

A3: Full Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

A3: Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.4692	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.2037	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.8431	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.2872	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.1426	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Essay Economics Terms (%)	1.0382	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
Elite U.S. University	1.0086	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
Letters Terms (%) PC1	0.8868	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018
TOEFL/IELTS	0.7541	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294

» Deviation Gains Attribution

A3: Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.0176	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	1.9874	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	1.8659	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.2973	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.0726	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.0219	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374

» Deviation Gains Attribution (Matriculation)

A3: Deviation Gains Robustness

	Acceptance Set	Matriculation Set
Observed Gain, Uncorrected	0.1015	0.1124
Observed Gain, Corrected	0.0514	0.0475
Perturbed Gain, Uncorrected	0.0937	0.1031
Perturbed Gain, Corrected	0.0435	0.0382
Likelihood-Ratio P-Value	0.1166	0.1166
Calibrated Signal Scale	0.4197	0.4197
Calibrated In-Sample Gain	0.1242	0.1481
Calibrated True Gain	0.0491	0.0560

» Deviation Gains (Acceptance)

» Deviation Gains (Matriculation)

- 1 Scale down (x, z) parameters with values of scaling factor τ from 0 to 1
- 2 Simulate *Success*, and find specific τ yielding observed joint LR test statistic
- 3 Calculate true deviation gain for that τ

A4: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

► Insufficiency of Comparing AMEs

A4: Graduate School Logistic Regression Results

Dependent Variable ← Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5752* (0.3479)	-0.5501 (0.4129)	-0.1766 (0.5545)
Winsorized Age	-0.0095*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0890)
Female	0.4079** (0.1681)	0.6723*** (0.2299)	0.6539** (0.2671)	0.1311 (0.3452)	-0.5060 (0.5606)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1611)
U.S. Asian	-0.5304 (0.4199)	-0.5669 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)
International Asian	-0.8445*** (0.2945)	0.1353 (0.4146)	0.9405 (0.7233)	0.4990 (0.6729)	-0.7827 (0.9323)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3523*** (0.5086)	-0.1061 (0.7483)	0.6239 (1.1070)
GRE Verbal Percentile	0.0155*** (0.0060)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.0607*** (0.0218)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0210* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)
Undergraduate GPA	2.2212*** (0.7219)	0.6968 (0.5476)	1.3648** (0.5513)	1.2545 (0.9703)	1.7772* (0.9487)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5456)	-0.4656 (0.6112)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3136)	0.6541* (0.3022)	-0.6057 (0.4065)	1.4303** (0.6102)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6276* (0.3754)	1.0690** (0.5390)	-0.6088 (1.4186)	2.3151** (0.6422)
Elite Foreign University	1.0061* (0.6060)	0.4750 (0.8224)	1.7715** (0.7938)	1.6195* (0.8587)	0.3889 (1.9889)
Other Foreign University	0.5442 (0.5765)	0.5777 (0.7686)	1.3732** (0.6902)	0.4188 (0.7157)	-1.0737 (1.3016)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4049 (0.4806)	0.4892 (0.6399)
Work Experience	0.3932** (0.1880)	0.4027 (0.2993)	0.4972 (0.3295)	0.1495 (0.3778)	1.8201** (0.8564)
#Professor Recommenders	0.0340 (0.1317)	-0.0179 (0.1339)	0.5511** (0.2554)	0.4781* (0.2662)	-0.0981 (0.2660)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5901* (0.3145)
#Math Courses	0.1809*** (0.0596)				
Advanced Math	0.2319 (0.1790)				
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3109)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)
Missing Undergraduate GPA	-0.4745 (0.4801)	-0.4917 (0.5440)	-1.5402** (0.6893)	-1.1097* (0.6481)	0.4660 (1.1831)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224* (0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2677	0.1525	0.1862	0.1871	0.2475

H robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303** (0.0153)	-0.0164 (0.0100)	-0.0431* (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	-0.0076*** (0.0027)	0.0017* (0.0010)	0.0028 (0.0025)	0.0012 (0.0029)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.0312** (0.0126)	0.0254*** (0.0088)	0.0490** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0229)	0.0207*** (0.0075)
U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0869*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0499 (0.0313)	0.0062 (0.0185)
International Asian	-0.0649*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.0385 (0.0243)	0.0143 (0.0157)	0.1013*** (0.0385)	-0.0057 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0026)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.0118*** (0.0014)	0.0006* (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0006** (0.0003)
GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	0.0009 (0.0007)	-0.0009 (0.0009)	0.0007** (0.0003)
Undergraduate GPA	0.1609*** (0.0551)	0.0264 (0.0208)	0.1022** (0.0423)	0.0484 (0.0522)	0.0757* (0.0418)	0.0488*** (0.0176)
TOEFL/IELTS	0.0297 (0.0219)	0.0244 (0.0179)	0.0821** (0.0417)	0.0523* (0.0301)	-0.0198 (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.0677*** (0.0225)	0.0149 (0.0119)	0.0490* (0.0281)	-0.0328 (0.0264)	0.0630** (0.0271)	0.0233** (0.0090)
Elite U.S. Liberal Arts College	0.0855*** (0.0288)	0.0238* (0.0143)	0.0801** (0.0402)	-0.0330 (0.0761)	0.0987** (0.0480)	0.0429** (0.0136)
Elite Foreign University	0.0770* (0.0463)	0.0180 (0.0311)	0.1327** (0.0591)	0.0877* (0.0481)	0.0166 (0.0850)	0.0479** (0.0241)
Other Foreign University	0.0416 (0.0441)	0.0219 (0.0201)	0.1026* (0.0527)	0.0227 (0.0391)	0.0468 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0036 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0246)	0.0061 (0.0205)	0.0776** (0.0368)	0.0221** (0.0099)
#Professor Recommenders	0.0026 (0.0101)	-0.0007 (0.0051)	0.0413** (0.0190)	0.0259* (0.0142)	-0.0042 (0.0113)	0.0003 (0.0047)
#Profic Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.0177 (0.0137)					
Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0491* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0206)	-0.1169*** (0.0524)	-0.0601** (0.0355)	0.0200 (0.0505)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0215)	0.0505 (0.0388)	-0.0024 (0.0276)	-0.0734* (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0323 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.5023 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4303)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7934 (1.1675)	-0.4305 (0.7939)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4965 (0.4275)	0.1866 (0.6866)
International Other	-0.1231 (0.2311)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807 (0.7583)
GRE Verbal Percentile	0.0167*** (0.0046)	0.0247** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0099)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0059)	-0.0073 (0.0062)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4085)	1.3552** (0.5706)	-1.3171** (0.5584)	-0.1802 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4762)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1759 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.8652 (0.7261)
Elite Foreign University	-0.1622 (0.3015)	-1.1920 (0.7484)	0.7404 (0.5668)	0.4801 (0.7841)	3.0530*** (0.8673)
Other Foreign University	-0.1071 (0.1979)	0.2042 (0.2732)	0.2721 (0.3301)	0.1731 (0.4202)	0.1731 (0.5926)
Graduate Degree	-0.0198 (0.1624)	0.1770 (0.2439)	-0.1601 (0.2765)	-0.8608*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4805)
@Professor Recommenders	-0.1059 (0.0909)	0.0011 (0.1138)	0.1828 (0.1473)	0.2181 (0.1817)	-0.4923** (0.1956)
@Prolific Recommenders	0.0039 (0.0956)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
@Math Courses	0.0366 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1993 (0.4328)	-0.2020 (0.2195)	-0.0011 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4817)
Missing Undergraduate GPA	0.1531 (0.1548)	0.0895 (0.2291)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2950* (0.1640)	0.4898 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-2.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.0524	0.0396	0.0753	0.1046	0.2230

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0291)	-0.0298 (0.0216)	-0.0345 (0.0262)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0008)	0.0054* (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0008)
Female	0.0108 (0.0143)	0.0041 (0.0239)	0.0131 (0.0294)	0.0162 (0.0251)	0.0123 (0.0078)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0306)	-0.0137 (0.0147)
U.S. Asian	0.0579 (0.0372)	0.0268 (0.0199)	-0.0750 (0.1101)	-0.0268 (0.0495)	0.0413 (0.0516)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0627 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0006)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019*** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281** (0.0553)	-0.0020** (0.0358)	-0.0058 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0496* (0.0261)	-0.0100 (0.0145)	0.0054 (0.0244)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0418)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0532 (0.0337)	0.0700 (0.0537)	0.0299 (0.0490)	0.1771*** (0.0484)	0.0158 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0018 (0.0321)	0.0015 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0151 (0.0261)	-0.0542*** (0.0206)	-0.0161 (0.0220)	-0.0093 (0.0085)
Work Experience	0.0258* (0.0148)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0171)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0286*** (0.0110)	-0.0027 (0.0042)
#Profic Recommenders	0.0004 (0.0067)	0.0008* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0128)	0.0083* (0.0045)
#Math Courses	0.0237 (0.0043)					
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0086)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0390 (0.0239)	-0.0098 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	0.0392* (0.0285)	-0.0207 (0.0201)	-0.0207 (0.0254)	0.0019 (0.0083)
Missing TOEFL/IELTS	-0.0302* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0084 (0.0272)	-0.1205*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R ²	0.2297	0.0912	0.0592	0.0850	0.0537

- Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

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